In the Name of God

Introduction to Machine Learning (25737-2)

Problem Set 6

Spring Semester 1403-04

Department of Electrical Engineering

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Due on 2 Tir, 1404 at 23:59

($*$) starred problems are optional!

1 Principle Component Analysis (PCA)

1.1 Basic PCA

Given 3 data points in 2-d space, $(1, 1)$, $(2, 2)$ and $(3, 3)$,

(a) what is the first principle component?

(b) If we want to project the original data points into 1-d space by principle component you choose, what is the variance of the projected data?

(c) For the projected data in (b), now if we represent them in the original 2-d space, what is the reconstruction error?

1.2 PCA and SVD

Given 6 data points in 5-d space, $(1, 1, 1, 0, 0)$, $(-3, -3, -3, 0, 0)$, $(2, 2, 2, 0, 0)$, $(0, 0, 0, -1, -1)$, $(0, 0, 0, 2, 2)$, $(0, 0, 0, -1, -1)$. We can represent these data points by a 6×5 matrix X, where each row corresponds to a data point:

$$X = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ -3 & -3 & -3 & 0 & 0 \\ 2 & 2 & 2 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & -1 & -1 \end{bmatrix}$$

(a) What is the sample mean of the data set?

(b) What is SVD of the data matrix X you choose?

Hint: The SVD for this matrix must take the following form, where $a, b, c, d, \sigma_1, \sigma_2$ are the parameters you need to decide.

$$X = \begin{bmatrix} a & 0 \\ -3a & 0 \\ 2a & 0 \\ 0 & b \\ 0 & -2b \\ 0 & b \end{bmatrix} \times \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} \times \begin{bmatrix} c & c & c & 0 & 0 \\ 0 & 0 & 0 & d & d \end{bmatrix}$$

(c) What is first principle component for the original data points?

(d) If we want to project the original data points into 1-d space by principle component you choose, what is the variance of the projected data?

(e) For the projected data in (d), now if we represent them in the original 5-d space, what is the reconstruction error?

Answer

Basic PCA

(a) The first principal component is the direction of maximum variance. For points (1,1), (2,2), (3,3) lying on the line $y = x$, the first principal component is $\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$.

(b) Projections onto the first principal component: $\begin{bmatrix} 1 \\ 1 \end{bmatrix} \to \sqrt{2}$, $\begin{bmatrix} 2 \\ 2 \end{bmatrix} \to 2\sqrt{2}$, $\begin{bmatrix} 3 \\ 3 \end{bmatrix} \to 3\sqrt{2}$.
Mean of projections: $2\sqrt{2}$.
Variance: $\frac{1}{3}\left[(\sqrt{2} - 2\sqrt{2})^2 + (2\sqrt{2} - 2\sqrt{2})^2 + (3\sqrt{2} - 2\sqrt{2})^2\right] = \frac{4}{3}$.

(c) Reconstructed points are identical to originals: (1,1), (2,2), (3,3). Reconstruction error is 0.

PCA and SVD

(a) Sample mean: $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ (data is centered).

(b) SVD parameters:
$a = \frac{1}{\sqrt{14}}$, $b = -\frac{1}{\sqrt{6}}$, $c = \frac{1}{\sqrt{3}}$, $d = \frac{1}{\sqrt{2}}$,
$\sigma_1 = \sqrt{42}$, $\sigma_2 = 2\sqrt{3}$.
Full SVD:

$$X = \begin{bmatrix} \frac{1}{\sqrt{14}} & 0 \\ -\frac{3}{\sqrt{14}} & 0 \\ \frac{2}{\sqrt{14}} & 0 \\ 0 & -\frac{1}{\sqrt{6}} \\ 0 & \frac{2}{\sqrt{6}} \\ 0 & -\frac{1}{\sqrt{6}} \end{bmatrix} \times \begin{bmatrix} \sqrt{42} & 0 \\ 0 & 2\sqrt{3} \end{bmatrix} \times \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

(c) First principal component (first column of V): $\begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & 0 & 0 \end{bmatrix}^\top$.

(d) Projections onto first principal component:
$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \end{bmatrix} \to \sqrt{3}$, $\begin{bmatrix} -3 & -3 & -3 & 0 & 0 \end{bmatrix} \to -3\sqrt{3}$,
$\begin{bmatrix} 2 & 2 & 2 & 0 & 0 \end{bmatrix} \to 2\sqrt{3}$, Other points $\to 0$.
Variance: $\frac{1}{6}\left[(\sqrt{3})^2 + (-3\sqrt{3})^2 + (2\sqrt{3})^2\right] = 7$.

(e) Reconstruction error:
First three points: perfect reconstruction (error 0).
Last three points: reconstructed as $\begin{bmatrix}0 & 0 & 0 & 0 & 0\end{bmatrix}$.
Errors: $0^2 + 0^2 + 0^2 + (-1)^2 + (-1)^2 = 2$ (4th),
$0^2 + 0^2 + 0^2 + 2^2 + 2^2 = 8$ (5th),
$0^2 + 0^2 + 0^2 + (-1)^2 + (-1)^2 = 2$ (6th).
Total reconstruction error: $0 + 0 + 0 + 2 + 8 + 2 = 12$.

2 Expectation Maximization

Imagine a machine learning class where the probability that a student gets an "A" grade is $\mathbb{P}(A) = 1/2$, a "B" grade $\mathbb{P}(B) = \mu$, a "C" grade $\mathbb{P}(C) = 2\mu$, and a "D" grade $\mathbb{P}(D) = 1/2 - 3\mu$. We are told that c students get a "C" and d students get a "D". We don't know how many students got exactly an "A" or exactly a "B". But we do know that h students got either an A or B. Therefore, a and b are unknown values where $a + b = h$. Our goal is to use expectation maximization to obtain a maximum likelihood estimate of μ.

1. Expectation step: Which formulas compute the expected values of a and b given μ? Circle your answers.

$$\hat{a} = \frac{1/2}{1/2 + h}\mu \quad \hat{b} = \frac{\mu}{1/2 + h}\mu$$

$$\hat{a} = \frac{1/2}{1/2 + \mu}h \quad \hat{b} = \frac{\mu}{1/2 + \mu}h$$

$$\hat{a} = \frac{\mu}{1/2 + \mu}h \quad \hat{b} = \frac{1/2}{1/2 + \mu}h$$

$$\hat{a} = \frac{1/2}{1 + \mu^2}h \quad \hat{b} = \frac{\mu}{1 + \mu^2}h$$

2. Maximization step: Given the expected values of a and b which formula computes the maximum likelihood estimate of μ? Circle your answer. *Hint:* Compute the MLE of μ assuming unobserved variables are replaced by their expectation.

$$\hat{\mu} = \frac{h - a + c}{6(h - a + c + d)}$$

$$\hat{\mu} = \frac{h - a + d}{6(h - 2a - d)}$$

$$\hat{\mu} = \frac{h - a}{6(h - 2a + c)}$$

$$\hat{\mu} = \frac{2(h - a)}{3(h - a + c + d)}$$

3. (True/False) Iterating between the E-step and M-step will *always* converge to a local optimum of μ (which may or may not also be a global optimum)? Explain in 1-2 sentences.

Answer

1. The correct formulas for the E-step are:

$$\boxed{\hat{a} = \frac{1/2}{1/2+\mu}h \quad \hat{b} = \frac{\mu}{1/2+\mu}h}$$

   Reason: These formulas compute the expected counts using conditional probabilities: $\hat{a} = h \cdot P(A \mid A \cup B) = h \cdot \frac{P(A)}{P(A)+P(B)} = h \cdot \frac{1/2}{1/2+\mu}$ $\hat{b} = h \cdot P(B \mid A \cup B) = h \cdot \frac{\mu}{1/2+\mu}$

2. The correct formula for the M-step is:

$$\boxed{\hat{\mu} = \frac{h-a+c}{6(h-a+c+d)}}$$

   Reason: Using the complete-data log-likelihood with expectations $\hat{a}, \hat{b}$: $\hat{\mu} = \frac{\hat{b}+c}{6(\hat{b}+c+d)}$. Substitute $\hat{b} = h - \hat{a}$ to match the option.

3. **True**. **Reason:** The EM algorithm guarantees monotonic increase in the log-likelihood, converging to a local optimum (or saddle point) of μ. This may not be the global optimum due to non-convexity of the likelihood function.

3 Dimension Reduction

(a) In linear PCA, the covariance matrix of the data $\mathbf{C} = \mathbf{X}^T\mathbf{X}$ is decomposed into weighted sums of its eigenvalues (λ) and eigenvectors $\mathbf{p}$:

$$\mathbf{C} = \sum_i \lambda_i \mathbf{p}_i \mathbf{p}_i^T$$

Prove mathematically that the first eigenvalue λ_1 is identical to the variance obtained by projecting data into the first principal component p_1.

Hint: PCA maximizes variance by projecting data onto its principal components.

(b) The key assumption of a naive Bayes (NB) classifier is that features are independent, which is not always desirable. Suppose that linear principal components analysis (PCA) is first used to transform the features, and NB is then used to classify data in this low-dimensional space. Is the following statement true? Justify your answers.

"The independent assumption of NB would now be valid with PCA transformed features because all principal components are orthogonal and hence uncorrelated."

Answer

a)

The variance of the projected data onto the first principal component $\mathbf{p}_1$ is defined as:

$$\text{Var(projection)} = \frac{1}{n-1}(\mathbf{X}\mathbf{p}_1)^\top(\mathbf{X}\mathbf{p}_1)$$

where $\mathbf{X}$ is the $n \times d$ data matrix (centered), and n is the number of samples. Given the covariance matrix $\mathbf{C} = \mathbf{X}^\top\mathbf{X}$, we have:

$$(\mathbf{X}\mathbf{p}_1)^\top(\mathbf{X}\mathbf{p}_1) = \mathbf{p}_1^\top(\mathbf{X}^\top\mathbf{X})\mathbf{p}_1 = \mathbf{p}_1^\top\mathbf{C}\mathbf{p}_1$$

Since $\mathbf{p}_1$ is an eigenvector of $\mathbf{C}$ with eigenvalue λ_1, it satisfies $\mathbf{C}\mathbf{p}_1 = \lambda_1 \mathbf{p}_1$. Thus:

$$\mathbf{p}_1^\top \mathbf{C}\mathbf{p}_1 = \mathbf{p}_1^\top(\lambda_1 \mathbf{p}_1) = \lambda_1(\mathbf{p}_1^\top \mathbf{p}_1) = \lambda_1 \cdot 1 = \lambda_1$$

Substituting back into the variance expression:

$$\text{Var(projection)} = \frac{1}{n-1}\lambda_1$$

The eigenvalue λ_1 of $\mathbf{C} = \mathbf{X}^\top \mathbf{X}$ is related to the sample covariance matrix $\mathbf{S} = \frac{1}{n-1}\mathbf{X}^\top \mathbf{X}$, which has eigenvalues $\frac{\lambda_i}{n-1}$. Therefore, $\lambda_1 = (n-1) \cdot \text{Var(projection)}$, meaning λ_1 equals $(n-1)$ times the variance of the projection. In the context of PCA optimization (maximizing scatter), λ_1 represents the **total scatter** (un-normalized variance) of the projected data, confirming λ_1 is identical to the scatter-based variance.

b)

The statement is **false**. Justification: Principal components (PCs) are orthogonal by construction, making them **uncorrelated** (linear independence). However, uncorrelatedness does not guarantee **statistical independence**, which is a stronger condition required by Naive Bayes. Statistical independence means the joint probability distribution factorizes as $P(\text{PC}_1, \text{PC}_2) = P(\text{PC}_1)P(\text{PC}_2)$, whereas uncorrelatedness only implies $E[\text{PC}_1\text{PC}_2] = E[\text{PC}_1]E[\text{PC}_2]$. Higher-order dependencies (e.g., quadratic, cubic) may still exist between PCs.

Further, PCA is **unsupervised** and ignores class labels. The transformation optimizes for reconstruction error, not conditional independence given the class label. Thus, PCs may exhibit dependencies *within* classes even if globally uncorrelated. For example, if two PCs satisfy $\text{PC}_2 = \text{PC}_1^2$ (a nonlinear dependency), they are uncorrelated but not independent, violating Naive Bayes's assumption.

4 PCA Fundamentals

Principal Component Analysis (PCA) is a commonly used technique for dimensionality reduction. Suppose we are given a set of data points $\mathcal{D} = \{x_n \in \mathbb{R}^D : n = 1, ..., N\}$. PCA can be viewed as a change of basis that represents each data point x_n as a weighted combination of a new set of basis functions $w_1, ..., w_L \in \mathbb{R}^D$, where the weights are given by $z_n \in \mathbb{R}^L$. That is, each x_n is approximated by $\sum_{k=1}^{L} z_{nk} w_k$. Here we explore one of the many aspects of this change of basis operation.

(a) Show that the sample covariance matrix Σ is PSD, where

$$\Sigma = \frac{1}{N}\sum_{n=1}^{N}(x_n - \mu)(x_n - \mu)^T$$

and $\mu = \frac{1}{N}\sum_{n=1}^{N} x_n$ is the sample mean.

(b) One way to view the change of basis that PCA performs is that it is the optimal solution that minimizes the average reconstruction error of the data:

$$\mathcal{L}(W, Z) = \frac{1}{N}\|X - ZW^T\|_F^2 = \frac{1}{N}\sum_{n=1}^{N}\|x_n - Wz_n\|^2$$

where $X \in \mathbb{R}^{N\times D}$ is the data matrix, $W \in \mathbb{R}^{D\times L}$ is the matrix of "latent factors", and $Z \in \mathbb{R}^{N\times L}$ is the matrix of "latent vectors".

Suppose we wish to minimize the error, subject to the constraint that W is an orthogonal matrix. We will first find the optimal Z and then find the optimal W when $L = 1$. Express and expand out the reconstruction error as a function of w_1 and its associated coefficients $z_1 = [z_{11}, ..., z_{N1}] \in \mathbb{R}^N$, and show the optimal solution for each entry of z_1, z_{n1}, is $z_{n1} = w_1^T x_n$.

(c) Given the optimal solution of z_n, show that the optimal solution for w_1 corresponds to finding the largest eigenvalue of some matrix. What is the matrix in the case in which the data is centered, i.e., the sample mean $\mu = 0$? **Hint**: This will require using constrained optimization.

(d) It is often said that the optimal solution for w_1 that minimizes the reconstruction error corresponds to maximizing the sample variance of the projected data. Show that this is the case when the data is centered. Give a brief explanation for what might go wrong when the data is not centered.

Answer

a)

The sample covariance matrix Σ is positive semi-definite (PSD). For any non-zero vector $v \in \mathbb{R}^D$:

$$v^T \Sigma v = \frac{1}{N} \sum_{n=1}^{N} v^T (x_n - \mu)(x_n - \mu)^T v = \frac{1}{N} \sum_{n=1}^{N} \left[(x_n - \mu)^T v\right]^2 \geq 0$$

This shows Σ is PSD as the expression is always non-negative.

b)

The reconstruction error for $L = 1$ is:

$$\mathcal{L}(w_1, z_1) = \frac{1}{N} \sum_{n=1}^{N} \|x_n - z_{n1} w_1\|^2$$

Expanding the norm:

$$= \frac{1}{N} \sum_{n=1}^{N} \left(x_n^T x_n - 2 z_{n1} w_1^T x_n + z_{n1}^2 w_1^T w_1\right)$$

Since W is orthogonal, $\|w_1\| = 1$ ($w_1^T w_1 = 1$). Taking derivative with respect to z_{n1}:

$$\frac{\partial \mathcal{L}}{\partial z_{n1}} = \frac{1}{N} \left(-2 w_1^T x_n + 2 z_{n1}\right) = 0$$

Solving yields:

$$z_{n1} = w_1^T x_n$$

c)

Substituting $z_{n1} = w_1^T x_n$ into the loss function:

$$\mathcal{L}(w_1) = \frac{1}{N} \sum_{n=1}^{N} \left(\|x_n\|^2 - (w_1^T x_n)^2\right)$$

Minimizing $\mathcal{L}$ is equivalent to maximizing $\frac{1}{N} \sum_{n=1}^{N} (w_1^T x_n)^2$. For centered data ($\mu = 0$), this equals $w_1^T \left(\frac{1}{N} \sum_{n=1}^{N} x_n x_n^T\right) w_1 = w_1^T \Sigma w_1$. Subject to $\|w_1\| = 1$, the solution is the eigenvector of Σ (sample covariance matrix) corresponding to its largest eigenvalue.

d)

For centered data, the sample variance of projected data is:

$$\mathrm{Var}(z) = \frac{1}{N}\sum_{n=1}^{N}(w_1^T x_n)^2 = w_1^T \Sigma w_1$$

which is exactly the term maximized in (c). When data is not centered, the projection $z_{n1} = w_1^T x_n$ includes the mean, and maximizing $\sum(w_1^T x_n)^2$ may prioritize capturing the mean rather than the direction of maximum variance, potentially misaligning with the true principal component.

5 Learning Mixtures of Gaussians with k-Means

Let $X_1, \ldots, X_n \in \mathbb{R}^d$ be independent, identically distributed points sampled from a mixture of two normal (Gaussian) distributions. They are drawn independently from the probability distribution function (PDF)

$$p(x) = \theta N_1(x) + (1-\theta)N_2(x), \quad \text{where } N_1(x) = \frac{1}{(\sqrt{2\pi})^d}e^{-\|x-\mu_1\|^2/2} \text{ and } N_2(x) = \frac{1}{(\sqrt{2\pi})^d}e^{-\|x-\mu_2\|^2/2}$$

are the PDFs for the isotropic multivariate normal distributions $\mathcal{N}(\mu_1, 1)$ and $\mathcal{N}(\mu_2, 1)$, respectively. The parameter $\theta \in (0, 1)$ is called the mixture proportion. In essence, we flip a biased coin to decide whether to draw a point from the first Gaussian (with probability θ) or the second (with probability $1 - \theta$).

Each data point is generated as follows. First draw a random Z_i, which has value 1 with probability θ, and has value 2 with probability $1-\theta$. Then, draw $X_i \sim \mathcal{N}(\mu_{Z_i}, 1)$. Our learning algorithm gets X_i as an input, but does not know Z_i.

Our goal is to find the maximum likelihood estimates of the three unknown distribution parameters $\theta \in (0, 1)$, $\mu_1 \in \mathbb{R}^d$, and $\mu_2 \in \mathbb{R}^d$ from the sample points $X_1, \ldots, X_n$. Unlike MLE for one Gaussian, it is **not** possible to give explicit analytic formulas for these estimates. Instead, we develop a variant of k-means clustering which (often) converges to the correct maximum likelihood estimates of θ, μ_1, μ_2. This variant doesn't assign each point entirely to one cluster; rather, each point is assigned an estimated posterior probability of coming from normal distribution 1.

(a) Let $\tau_i = P(Z_i = 1|X_i)$. That is, τ_i is the posterior probability that point X_i has $Z_i = 1$. Use Bayes' Theorem to express τ_i in terms of $X_i, \theta, \mu_1, \mu_2$, and the Gaussian PDFs $N_1(x)$ and $N_2(x)$. To help you with part (c), also write down a similar formula for $1 - \tau_i$, which is the posterior probability that $Z_i = 2$.

(b) Write down the log-likelihood function, $\ell(\theta, \mu_1, \mu_2; X_1, \ldots, X_n) = \ln p(X_1, \ldots, X_n)$, as a summation. Note: it doesn't simplify much.

(c) Express $\frac{\partial \ell}{\partial \theta}$ in terms of θ and τ_i, $i \in \{1, \ldots, n\}$ and simplify as much as possible. There should be no normal PDFs explicitly in your solution, though the τ_i's may implicitly use them.

Hint: Recall that: $\ln f(x))' = \frac{f'(x)}{f(x)}$.

Answer

a)

By Bayes' theorem:

$$\tau_i = P(Z_i = 1|X_i) = \frac{P(Z_i = 1)P(X_i|Z_i = 1)}{P(X_i)} = \frac{\theta \cdot N_1(X_i)}{\theta \cdot N_1(X_i) + (1-\theta) \cdot N_2(X_i)}$$

$$1 - \tau_i = P(Z_i = 2|X_i) = \frac{P(Z_i = 2)P(X_i|Z_i = 2)}{P(X_i)} = \frac{(1-\theta) \cdot N_2(X_i)}{\theta \cdot N_1(X_i) + (1-\theta) \cdot N_2(X_i)}$$

b)

The log-likelihood function is:

$$\ell(\theta, \mu_1, \mu_2; X_1, \ldots, X_n) = \sum_{i=1}^{n} \ln\left(\theta N_1(X_i) + (1-\theta) N_2(X_i)\right)$$

c)

The partial derivative with respect to θ is:

$$\frac{\partial \ell}{\partial \theta} = \sum_{i=1}^{n} \left(\frac{\tau_i}{\theta} - \frac{1-\tau_i}{1-\theta} \right) = \frac{1}{\theta(1-\theta)} \sum_{i=1}^{n} (\tau_i - \theta)$$

This simplifies to:

$$\frac{\partial \ell}{\partial \theta} = \frac{\sum_{i=1}^{n} \tau_i - n\theta}{\theta(1-\theta)}$$

6 Random Projection

Recall the method of random projection, where we project every sample point $x \in \mathbb{R}^d$ onto a randomly chosen k-dimensional subspace $S \subset \mathbb{R}^d$, where $k < d$. Let's consider a closely related method for dimensionality reduction that also approximately preserves distances between points (with high probability).

(a) Recall from our discussion of principal components analysis (PCA) that, if two conditions hold, we can orthogonally project a point $x \in \mathbb{R}^d$ onto the subspace spanned by the vectors $v_1, \ldots, v_k$ with the formula $\tilde{x} = \sum_{i=1}^{k} (x \cdot v_i) v_i$. What two conditions must $v_1, \ldots, v_k$ satisfy for this formula to be correct?

(b) Consider a random matrix $G \in \mathbb{R}^{k \times d}$, with $k < d$. Every component of G is a random real number $G_{ij} \sim \mathcal{N}(0, 1/d)$; that is, every component is drawn from a univariate normal distribution with mean zero and variance $1/d$. These components are all independent of each other. Consider a (not random) vector $x \in \mathbb{R}^d$. Show that the expected squared ℓ_2-norm of Gx is

$$E\left[||Gx||^2\right] = \frac{k}{d}||x||^2.$$

(c) * The result in part (b) suggests that if we choose a random G as described in part (b) and replace the sample points $X_1, X_2, \ldots, X_n \in \mathbb{R}^d$ with the k-dimensional points

$$\tilde{X}_1 = \sqrt{\frac{d}{k}} G X_1, \quad \tilde{X}_2 = \sqrt{\frac{d}{k}} G X_2, \quad \ldots, \quad \tilde{X}_n = \sqrt{\frac{d}{k}} G X_n,$$

The distance between two points might not "change much"; that is, $||\tilde{X}_i - \tilde{X}_j||$ is an approximation of $||X_i - X_j||$. This is similar to computing an orthogonal projection onto a randomly chosen k-dimensional subspace, but not quite the same.

In the limit as $d \to \infty$ (with k fixed), why does this method become functionally the same as the method of random projection?

Hint: What two counterintuitive properties of vectors in high-dimensional spaces are relevant, and which vectors should we apply them to? (A written answer suffices; no math is necessary.)

Answer

a)

The two conditions for orthogonal projection are:

1. The vectors $v_1, \ldots, v_k$ must be **unit vectors**: $\|v_i\|_2 = 1$ for all i
2. The vectors must be **pairwise orthogonal**: $v_i^T v_j = 0$ for all $i \neq j$

b)

The expected squared ℓ_2-norm is:

$$\mathbb{E}\left[\|Gx\|^2\right] = \mathbb{E}\left[\sum_{i=1}^{k} (Gx)_i^2\right] = \sum_{i=1}^{k} \mathbb{E}\left[\left(\sum_{j=1}^{d} G_{ij} x_j\right)^2\right]$$

Since $G_{ij} \sim \mathcal{N}(0, 1/d)$ are independent:

$$\mathbb{E}\left[\left(\sum_{j=1}^{d} G_{ij} x_j\right)^2\right] = \mathrm{Var}\left(\sum_{j=1}^{d} G_{ij} x_j\right) = \sum_{j=1}^{d} \mathrm{Var}(G_{ij}) x_j^2 = \sum_{j=1}^{d} \frac{1}{d} x_j^2 = \frac{1}{d}\|x\|^2$$

Thus:

$$\mathbb{E}\left[\|Gx\|^2\right] = \sum_{i=1}^{k} \frac{1}{d}\|x\|^2 = \frac{k}{d}\|x\|^2$$

c)

In high-dimensional spaces ($d \to \infty$ with k fixed):

1. The rows of G become **approximately orthogonal**: $g_i \cdot g_j \to 0$ for $i \neq j$
2. The norms of the rows converge to 1: $\|g_i\|^2 \to 1$ for all i

This makes G behave like an orthogonal projection matrix (up to scaling), as the row space forms an orthonormal basis for a random k-dimensional subspace.

7 * Kernel Principal Components Analysis

In this problem, we will derive the kernelized version of principal components analysis (PCA). You are given a **centered** $n \times d$ design matrix X whose rows express the sample points $X_1^T, X_2^T, ..., X_n^T$, and a feature map Φ that maps each sample point $X_i \in \mathbb{R}^d$ to a high-dimensional "lifted" sample point $\Phi(X_i) \in \mathbb{R}^D$, where $D \gg d$. Define the kernel function $k(X_i, X_j) = \Phi(X_i)^T \Phi(X_j)$, and suppose it takes $O(d)$ time to compute (which is much, much less than D). The sample covariance matrix of the lifted points in $\mathbb{R}^D$ is:

$$C = \frac{1}{n}\Phi(X)^T\Phi(X) = \frac{1}{n}\sum_{i=1}^{n}\Phi(X_i)\Phi(X_i)^T \in \mathbb{R}^{D\times D}$$

Recall that $\Phi(X) \in \mathbb{R}^{n\times D}$. Recall also that our convention is to express each X_i and each $\Phi(X_i)$ as a column vector, but these points are stored as rows of X and $\Phi(X)$, respectively, which is why the transpose moved.

(a) To perform PCA, we want to find the eigenvectors of C with nonzero eigenvalues. However, we don't ever want to construct C—it's way too big! We don't even want to construct one eigenvector of C? Show that if a vector $v \in \mathbb{R}^D$ is an eigenvector of C with a nonzero eigenvalue, then v can be expressed as a linear combination of lifted sample points: that is, $v = \sum_{j=1}^{n} a_j \Phi(X_j) = \Phi(X)^T a$ for some vector $a \in \mathbb{R}^n$ of "dual" coefficients. *Hint:* think of the standard definition of an eigenvector of C, then substitute in the value of C.

(b) Recall that the kernel matrix $K = \Phi(X)\Phi(X)^T$ is the symmetric $n \times n$ matrix such that $K_{ij} = k(X_i, X_j)$. In the circumstance described in part (a), prove that $K^2 a = \lambda n K a$. *Hint:* again start with the standard definition of an eigenvector of C, then substitute in the value of v from above.

(c) Conversely, suppose that a is an eigenvector of K with eigenvalue λ_a. Show that $v = \Phi(X)^T a$ is an eigenvector of C. What is v's corresponding eigenvalue?

(d) The parts above imply that by finding the eigendecomposition of the kernel matrix K, we can indirectly express the eigenvectors (that have nonzero eigenvalues) of the kernelized covariance matrix C without ever explicitly computing C or any vector of length D. Suppose that we have a test point $z \in \mathbb{R}^d$ and we want to efficiently compute the principal coordinate of $\Phi(z)$ projected onto a principal component $v = \sum_{j=1}^{n} a_j \Phi(X_j) = \Phi(X)^T a$. Assume that we have already computed a. Describe how to compute $\Phi(z)$'s principal coordinate quickly, without ever computing a vector of length $\Theta(D)$ or doing a computation taking $\Theta(D)$ time. (Recall our assumption that kernel function computations are much, much faster than that.)

Answer

a)

Let v be an eigenvector of C with eigenvalue $\lambda \neq 0$:

$$Cv = \lambda v$$

Substitute $C = \frac{1}{n}\Phi(X)^T\Phi(X)$:

$$\frac{1}{n}\Phi(X)^T\Phi(X)v = \lambda v$$

Rearrange:

$$\Phi(X)^T\Phi(X)v = n\lambda v$$

$$\Phi(X)^T(\Phi(X)v) = n\lambda v$$

This shows v is in the column space of $\Phi(X)^T$, which is spanned by $\{\Phi(X_1), \dots, \Phi(X_n)\}$. Thus:

$$v = \sum_{j=1}^{n} a_j \Phi(X_j) = \Phi(X)^T a \quad \text{for some} \quad a \in \mathbb{R}^n$$

b)

Using $v = \Phi(X)^T a$ in $Cv = \lambda v$:

$$\frac{1}{n}\Phi(X)^T \Phi(X)(\Phi(X)^T a) = \lambda \Phi(X)^T a$$

Multiply both sides by $n\Phi(X)$:

$$\Phi(X)\Phi(X)^T \Phi(X)\Phi(X)^T a = n\lambda \Phi(X)\Phi(X)^T a$$

Recognize $K = \Phi(X)\Phi(X)^T$:

$$K^2 a = n\lambda K a$$

c)

Let a be an eigenvector of K with eigenvalue λ_a:

$$Ka = \lambda_a a$$

Form $v = \Phi(X)^T a$ and compute Cv:

$$Cv = \frac{1}{n}\Phi(X)^T \Phi(X)(\Phi(X)^T a) = \frac{1}{n}\Phi(X)^T K a = \frac{1}{n}\Phi(X)^T(\lambda_a a) = \frac{\lambda_a}{n} v$$

Thus v is an eigenvector of C with eigenvalue λ_a/n.

d)

The principal coordinate (projection of $\Phi(z)$ onto normalized v) is:

$$\text{proj}_v(\Phi(z)) = \frac{v^T \Phi(z)}{\|v\|}$$

Compute numerator:

$$v^T \Phi(z) = a^T \Phi(X)\Phi(z) = \sum_{j=1}^{n} a_j k(X_j, z)$$

Compute denominator:

$$\|v\|^2 = v^T v = a^T \Phi(X)\Phi(X)^T a = a^T K a = \lambda_a (a^T a)$$

$$\|v\| = \sqrt{\lambda_a (a^T a)}$$

Thus:

$$\text{proj}_v(\Phi(z)) = \frac{\sum_{j=1}^{n} a_j k(X_j, z)}{\sqrt{\lambda_a (a^T a)}}$$

This requires $O(nd)$ time (each $k(X_j, z)$ is $O(d)$, sum is $O(n)$), avoiding $\Theta(D)$ computations.